

EC8208 – Software Architecture
Continuous Assessments Structure

You need to form a group and select a real-world problem domain and propose a software solution.

Note :The same scenario must be used throughout all three parts of the assessment.

Continuous Assessment 01 – Architecture Analysis and Design Report

(20 Marks)

You are required to analyze a real-world problem and design an appropriate software architecture for the proposed solution.

Report Contents

1. Introduction

- Background of the selected problem
- Problem statement
- Objectives of the proposed system

2. Requirements Analysis

- Functional requirements
- Non-functional requirements
- Stakeholder identification

3. Architecture Selection

Students must:

- Identify a possible architecture styles

Examples:

4. Architectural Design

Include the following diagrams where applicable:

- Context Diagram
- Component Diagram
- Deployment Diagram
- Class Diagram
- Data Flow Diagram

- Architecture Diagram

5. Technology Stack Selection

6. Quality Attributes

Discuss how the design addresses:

- Scalability
- Performance
- Security
- Availability
- Maintainability
- Reliability

7. Conclusion

Summarize the proposed solution and architectural decisions.

Submission Requirements

- Report Length: 10–20 pages
- Font: Times New Roman, Size 12
- Line Spacing: 1.5
- Format: PDF

Note: This is a group take home assessment

Continuous assessment 02 – Individual Written Assessment

(10 Marks)

You will individually complete a written assessment based on the architecture proposed in Part 1.

Note: This is an individual assessment and contributes separately to the final grade.

Continuous assessment 03 – Prototype Development and Presentation

(20 Marks)

You must develop a functional prototype implementing the architecture proposed in Part 1.

The prototype should demonstrate the key architectural concepts and major system functionalities.

The prototype should include:

- Simple User Interface
- Business Logic Layer
- Data Storage Layer
- At least three major functionalities
- Error handling
- Basic security mechanisms
- Deployment demonstration

Deliverables

1. *Source Code*

2. *Prototype Demonstration*

- System functionality
- Architecture implementation
- Communication between components
- Key design decisions

3. *Presentation*

Each group will deliver a presentation covering:

- Problem overview
- Architecture selection
- Architectural diagrams
- Technology stack
- Prototype demonstration

Presentation Duration

- 10 minutes presentation
- 5 minutes Q&A session

Note : This is a group assessment and all team members must participate.